

Experiment 3

Designing a Quarter Wave Transformer by Varying Frequency, Input Impedance and Characteristic Impedance

Test Case 1

Design a quarter-wave transformer for an input impedance (Z_{in}) of $50\ \Omega$ and a load impedance (Z_L) of $100\ \Omega$. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

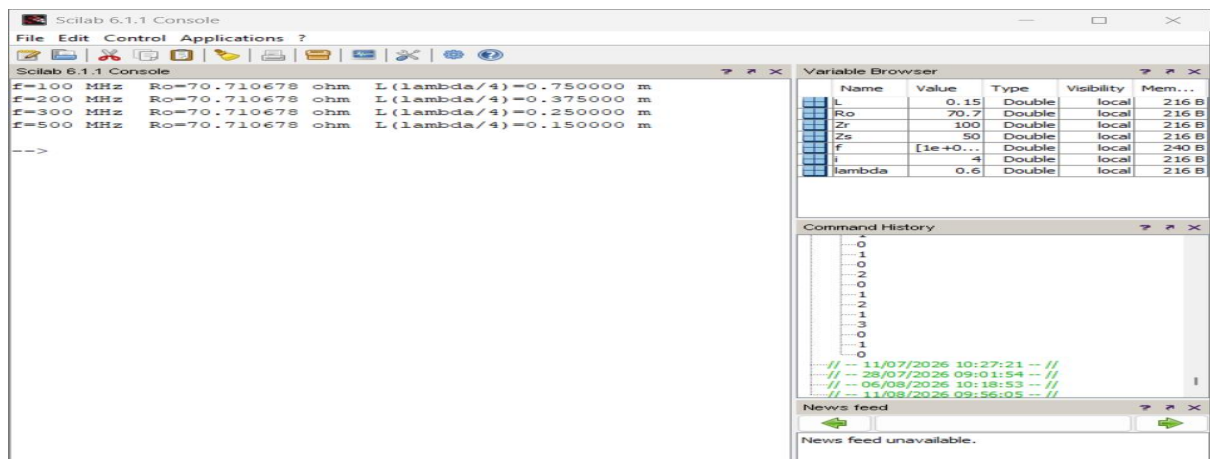
Code -

```
clear; clc;
Zr=100; Zs=50;
Ro=sqrt(Zs*Zr);
f=[100 200 300 500]*10^6;
for i=1:4
    lambda=300/(f(i)*10^-6);    // wavelength in metres
    L=lambda/4;                // quarter-wave length
    printf('f=%d MHz  Ro=%f ohm  L(lambda/4)=%f m\n',f(i)*10^-6,Ro,L);
end
```

Output:

$Z_R = 100 \, \Omega$, $Z_{in} = 50 \, \Omega \rightarrow R_o = 70.71 \, \Omega$ (constant, independent of frequency)

Frequency (MHz)	Wavelength λ - Sim (m)	Length of $\lambda/4$ transformer - Theory (m)
100	3.00	0.75
200	1.50	0.375
300	1.00	0.25
500	0.60	0.15



Test Case 2

For a fixed operating frequency of 2 GHz and a load impedance (Z_L) of $100 \, \Omega$, design a quarter-wave transformer and evaluate its performance as the input impedance (Z_{in}) varies through $\{25, 50, 75, 100\} \, \Omega$. Validate that $R_o = \sqrt{Z_{in} \cdot Z_L}$ changes with Z_{in} .

Code -

```
clear; clc;
ZL=100; f=2*10^9;
Zin=[25 50 75 100];
for i=1:4
```

```

Ro=sqrt(Zin(i)*ZL);
printf('Zin=%d ohm  Ro=%f ohm\n',Zin(i),Ro);
end

```

Output:

$ZL = 100 \, \Omega$, $f = 2 \, \text{GHz}$ → Variable: Zin

Zin (Ω)	Ro (ohms) - Simulation	Ro (ohms) - Theoretical
25	50.00	50.00
50	70.71	70.71
75	86.60	86.60
100	100.00	100.00

Scilab 6.1.1 Console

File Edit Control Applications ?

Scilab 6.1.1 Console

```

Zin=25 ohm  Ro=50.000000 ohm
Zin=50 ohm  Ro=70.710678 ohm
Zin=75 ohm  Ro=86.602540 ohm
Zin=100 ohm  Ro=100.000000 ohm
-->

```

Variable Browser

Name	Value	Type	Visibility	Mem...
Ro	100	Double	local	216 8
ZL	100	Double	local	216 8
Zin	[25, 5...	Double	local	240 8
f	2e+09	Double	local	216 8
i	4	Double	local	216 8

Command History

```

0
1
0
2
0
1
2
1
3
0
1
0

```

News feed

News feed unavailable.